

CTP REVIEW CENTER

2/F Espana Place, Espana Blvd., Sampaloc, Manila

MASTER PLUMBERS PRE-BOARD EXAMINATION (July 28, 2012)

1. What is the minimum diameter of water supply pipe for flush valve Urinal closet?
 - a. $\frac{1}{2}$ " dia.
 - b. $\frac{3}{4}$ " dia.
 - c. $\frac{3}{8}$ " dia.
 - d. 1" dia.
2. Under uniform rate of charges for Master Plumbers Professional Practice, How much is the fee for the first plumbing unit?
 - a. P 300
 - b. P 400
 - c. P 500
 - d. P 600
3. What is the jointing method of installation for PVC pipes?
 - a. By threading
 - b. By solvent cement
 - c. By heating
 - d. None of the above
4. What is the standard length of Concrete pipe?
 - a. 3 meters
 - b. 2 meters
 - c. 1 meter
 - d. 1.50 meters
5. What is the usual pipe size to receive waste from bath tub?
 - a. 32 mm
 - b. 40 mm
 - c. 38 mm
 - d. 50 mm
6. How much do you charge for the sign and seal of plumbing design with a cost of P 800,000.00?
 - a. 2 %
 - b. 2.5 %
 - c. 3 %
 - d. 3.5 %
7. If there is no water closet, what is the minimum diameter of a pipe?
 - a. 3" dia
 - b. 2" dia
 - c. 6" dia
 - d. 4" dia.
8. In the digestion chamber, what is the minimum height of the water level?
 - a. 1.20 meters
 - b. 1.00 meter
 - c. 1.50 meters
 - d. 1.80 meters
9. What is the minimum water inlet for a flush type watercloset?
 - a. $\frac{3}{4}$ " dia
 - b. $\frac{1}{2}$ " dia
 - c. 1" dia
 - d. $\frac{3}{8}$ " dia
10. At what temperature the sprinkler head automatically opens and emit a series of fine water spray?
 - a. 120 deg. F
 - b. 125 deg. F
 - c. 135 deg. F
 - d. 115 deg. F

11. What is the typical distance between pendant sprinkler heads on a commercial building?
- a. 3000 mm
 - b. 3800 mm
 - c. 3600 mm
 - d. 4000 mm
12. In the length of 7 meters of soil and waste Cast Iron pipe to be installed, what is the right quantity take off and specs of pipe?
- a. 4-single hub, 1-double hub
 - b. 4-single hub
 - c. 2-single hub, 2-double hub
 - d. 4-double hub
13. What is the best water heating system to be used for both lavatory and shower?
- a. Multipoint system
 - b. Centralized system
 - c. Single point system
 - d. None of the above
14. What kind of pump is suited to raise water for tall buildings?
- a. Piston pump
 - b. Centrifugal pump
 - c. Booster pump
 - d. None of the above
15. What is the height required for vent stack thru roof above the roof if the roof is use for some purpose like roof deck?
- a. 6 feet
 - b. 8 feet
 - c. 12 feet
 - d. 1 foot
16. What fitting is suited to receive discharge directly from water closet if the branch is right under it with ample space?
- a. Wye 4" x 4"
 - b. Tee 4" x 4"
 - c. Sanitary Tee 4" x 4"
 - d. Long sweep bend elbow
17. What kind of check valve to be used to prevent water from flowing back in a vertical position?
- a. Lift check valve
 - b. Horizontal check valve
 - c. Swing Check valve
 - d. Vertical check valve
18. What is the required height of waste stub-out for wall mount lavatory?
- a. 600 mm
 - b. 501 mm
 - c. 400 mm
 - d. 460 mm
19. What is the minimum height of water supply pipe measured from the top of urinal?
- a. 170 mm
 - b. 180 mm
 - c. 200 mm
 - d. 300 mm
20. What is the minimum height of water supply pipe stub-out of water closet with tank from finish floor?
- a. 158 mm
 - b. 114 mm
 - c. 200 mm
 - d. 150 mm
21. A water closet that is easily recognized by the bulge on front and the most noisiest of all types.

- a. Wash down type
 - b. Reverse trap type
 - c. Siphon action type
 - d. Siphon vortex type
22. What kind of pipe with hub from one end and spigot on the other end?
- a. PVC pipe
 - b. Cast Iron pipe
 - c. Concrete pipe
 - d. Asbestos pipe
23. What is the commercial length of Sanitary PVC pipes?
- a. 4 meters
 - b. 3 meters
 - c. 5 meters
 - d. 6 meters
24. What is the type of sewage disposal system that allows the raw contaminated sewage to leach into the underground surface?
- a. Septic Tank
 - b. Cesspool
 - c. Public Sewer line
 - d. Privy
25. What is the minimum requirement for a water closet in dormitories for 10 male?
- a. 1
 - b. 2
 - c. 3
 - d. 4
26. What kind of pipe is commonly use for gas?
- a. G.I. pipes
 - b. PVC pipes
 - c. PPR pipes
 - d. Copper pipes
27. In the plumbing plans, solid lines is use for the lines of?
- a. Soil and waste lines
 - b. Water lines
 - c. Drainage lines
 - d. Vents lines
28. What plumbing tools to be used in tightening the connections between G.I. pipes and fittings?
- a. Strap Wrench
 - b. Chain wrench
 - c. Pipe wrench
 - d. Pipe Threaders
29. What water pipes that is measured from its outside diameter?
- a. PPR pipes
 - b. G.I. pipes
 - c. PVC pipes
 - d. Copper pipes
30. What valve that controls water from back flowing?
- a. Gate valve
 - b. Check valve
 - c. Relief valve
 - d. None of the above
31. What particular PVC fitting that receives unit vent pipe, and two lavatories waste pipes?
- a. Sanitary cross tee
 - b. Double bend
 - c. Wye
 - d. Tee

Situation: II A 4-Storey Apartment building has a total of 16 water closets with tank, 16 shower head and valves, 16 lavatories, and 16 kitchen sinks, and 16 floor drains that connects to one soil stack with stack vent with 2 soil branch pipe in each floor.

32. What is the size of the supply pipe per apartment?
- $\frac{3}{4}$ " dia.
 - $\frac{1}{2}$ " dia.
 - 1" dia.
 - $\frac{3}{8}$ " dia.
33. How many angle valves to be use?
- 10 pcs.
 - 12 pcs.
 - 32 pcs.
 - 48 pcs.
34. In the given problem, what should be the size of the soil stack?
- 6" dia.
 - 5" dia.
 - 4" dia.
 - 3" dia.
35. In the situation given, how many 4" dia. clean out is required?
- 8 pcs.
 - 9 pcs
 - 10 pcs.
 - 12 pcs.
36. How many sanitary cross tee 4" x 4" dia. is required?
- 10 pcs
 - 16 pcs
 - 30 pcs.
 - 32 pcs.
37. How many $\frac{3}{8}$ " dia. flexible hose are needed?
- 14 pcs.
 - 16 pcs.
 - 18 pcs.
 - 32 pcs.
38. What is the size of the branch soil pipe in each floor?
- 3" dia.
 - 4" dia.
 - 5" dia.
 - 6" dia.
39. What is the size of the stack vent?
- 2" dia.
 - 3" dia.
 - 4" dia.
 - 5" dia.
40. What is the total fixture unit of the above problem?
- 190 F.U.
 - 186 F.U.
 - 192 F.U.
 - 182 F.U.
41. What is the size of the sanitary drain to receive discharge from the fixture units above?
- 3" dia
 - 4" dia.
 - 5" dia..
 - 6" dia.
42. In the given problem, how may Sanitary cross tee is required?
- 3 pcs
 - 4 pcs.
 - 5 pcs
 - 6 pcs
43. If the house sewer has a distance of 20 meters from the building line to the main sewer with a depth of 400 mm, what is the rise angle of the pipe?
- 2 %
 - 3%
 - 2.5 %
 - 4 %

44. In the plumbing drawings, what is this line (- - - -) stands for?
- a. Soil pipe
 - b. Hot water line
 - c. Cold Water line
 - d. Vent
45. Every what distance should the clean out be installed in a long soil branch pipe?
- a. 15'
 - b. 30'
 - c. 40'
 - d. 50'
46. What kind of trap is appropriate for bathtub?
- a. P-trap
 - b. Grease trap
 - c. Drum trap
 - d. House trap
47. What is the device that is scientifically designed to prevent gases from back Flow into the outlet of the fixture?
- a. Fixture trap
 - b. Check valve
 - c. Pressure relief valve
 - d. House trap
48. How large is the down spout required to drain a roof with an area of 60 s.m.?
- a. 50 mm dia.
 - b. 75 mm dia.
 - c. 100 mm dia.
 - d. 125 mm dia.
49. In the given problem above, what is the size of the branch vent?
- a. 2" dia.
 - b. 3" dia.
 - c. 4" dia.
 - d. 5" dia.
50. What is the size of waste pipe of the pedestal urinal?
- a. 1-1/2" dia
 - b. 1-1/4" dia
 - c. 2" dia.
 - d. 3" dia.
51. What is the sewage flow per person in gallons for a single family dwellings?
- a. 50 gals.
 - b. 60 gals.
 - c. 75 gals.
 - d. 40 gals.
52. What is the minimum width of a septic tank?
- a. 1300 mm
 - b. 1200 mm
 - c. 900 mm
 - d. 1500 mm
53. What is the ratio of the leaching chamber against digestion chamber?
- a. 2/3
 - b. 1/3
 - c. 1/4
 - d. 1/2
54. A wrench that is use for turning large pipe?
- a. Stillson wrench
 - b. Pipe wrench
 - c. Chain wrench
 - d. Strap wrench
55. A tool used to secure a pipe?
- a. Monkey wrench
 - b. Basin wrench
 - c. Yoke vise
 - d. Stillson Wrench

56. A plumbing tool used to tighten connections in deeply recessed places.
- a. Monkey wrench
 - b. Basin Wrench
 - c. Pipe wrench
 - d. Strap wrench
57. What is the maximum distance of a P-trap to vent pipe?
- a. 1200 mm
 - b. 1500 mm
 - c. 1800 mm
 - d. None of the above
58. A tool that grips object and being pulled in tension around it.
- a. Chain wrench
 - b. Strap wrench
 - c. Stillson wrench
 - d. Monkey wrench
59. A hand tool w/ adjustable jaws for turning nuts of varying sizes & used for gripping hexagonal nuts.
- a. Monkey wrench
 - b. Pipe wrench
 - c. Stillson Wrench
 - d. Basin wrench
60. A fitting that receives P-trap of lavatory?
- a. Tapped tee
 - b. Sanitary tee
 - c. Wye
 - d. Waste pipe
61. What lubricant oil is use in pipe threading?
- a. Cooking oil
 - b. Olive oil
 - c. Animal oil
 - d. Engine oil
62. What material that seals caulked joint?
- a. Epoxy
 - b. Oakum
 - c. Pig lead
 - d. Solvent cement
63. How many fixture units that maybe connected to a 125 dia . size of sanitary drain in a 2% slope?
- a. 84 F.U.
 - b. 162 F.U.
 - c. 200 F.U.
 - d. 300 F.U.
64. What is the total fixture units for a small dwelling house with 1-water closet, 1-shower, 1- lavatory, 1-floor drain, 3-faucets?
- a. 12 F.U.
 - b. 13 F.U.
 - c. 13.5 F.U.
 - d. 14 F.U.
65. A valve that prevent reversal flow in the line?
- a. Globe valve
 - b. Angle valve
 - c. Safety valve
 - d. Check valve
66. A valve that is use for compressed air lines and other lines with excessive pressure.
- a. Foot valve
 - b. Safety valve
 - c. Check valve
 - d. Globe valve

67. In the uniform rate of charges, how much is the charge per unit for the succeeding plumbing units to mean additional water closet, urinal and bathtub?
- a. P 200
 - b. P 250
 - c. P 300
 - d. P 500
68. It is a device that inserted into water main that serves as a control stop for water service?
- a. Corporation cock
 - b. Curb Stop
 - c. Safety valve
 - d. Meter stop
69. What is the minimum diameter of the storage tank?
- a. 30 cm
 - b. 50 cm
 - c. 60 cm
 - d. 130 cm
70. What is the standard metal gauge pressure of a storage tank?
- a. 60 psi
 - b. 75 psi
 - c. 85 psi
 - d. None of the above
71. What is the maximum length of a range boiler?
- a. 150 cm
 - b. 180 cm
 - c. 200 cm
 - d. 210 cm
72. In hot water consumption, what is the gallons per person per hour for residential use?
- a. 5 gals.
 - b. 3 gals.
 - c. 8 gals
 - d. 10 gals.
73. On average energy wasted on a leaking faucet, how many gallons of water are wasted in a 30 drops per minute?
- a. 54 gals
 - b. 60 gals
 - c. 65 gals
 - d. 72 gals
74. What is the mounting height of lavatory from toilet finish floor line?
- a. 787 mm
 - b. 800 mm
 - c. 820 mm
 - d. 750 mm
75. What kind of vent is use to ventilate two fixture traps that discharges into sanitary cross tee with deflector?
- a. Circuit vent
 - b. Relief vent
 - c. Unit vent
 - d. None of the above
76. What is the permissible number of fixture units allowed to a 3" diameter of pipe?
- a. 18 F.U.
 - b. 36 F.U.
 - c. 72 F.U.
 - d. 40 F.U.
77. A fitting that is necessary for maintenance and is required for every change of direction?
- a. P-trap
 - c. Wye

- b. Elbow
 - d. Clean out
78. In order to connect flexible hose, what is the kind of fitting to use?
- a. Tee
 - c. Nipple
 - b. Elbow
 - d. None of the above
79. What is the minimum diameter of a slop sink waste pipe?
- a. 38 mm dia
 - c. 75 mm dia
 - b. 32 mm dia
 - d. 50 mm dia.
80. How many gallons of water per minute a 20 mm diameter service pipe could deliver and sufficient enough to serve at least 10 fixtures?
- a. 15 gallons
 - c. 18 gallons
 - b. 20 gallons
 - d. 10 gallons
81. What is the color of Fire Protection pipes?
- a. Blue
 - c. Red
 - b. Black
 - d. White
82. What is the working pressure in a one piece Water closet (Tank and bowl)?
- a. 20-25 psi
 - c. 5-10 psi?
 - b. 30-35 psi
 - d. 10-15 psi?
83. Trap seal maybe lost due to?
- a. Capillary attraction
 - c. Evaporation
 - b. Wind effect
 - d. All of the above
84. How many oakum is needed in caulking a joint of 100 mm diameter cast iron pipe?
- a. 4 feet
 - c. 6 feet
 - b. 5 feet
 - d. 6 -1/2 feet
85. Anaerobic bacteria lives in ?
- a. Septic tank w/ oxygen
 - c. Raw sewage
 - b. Septic tank without oxygen
 - d. None of the above
86. What ventilation system is usually used for multi-storey building?
- a. Unit vent
 - c. Circuit vent
 - b. Wet vent
 - d. Back vent
87. What vent is needed if a horizontal branch pipe is close to public sewer, and has a sink with food grinder connected on it?
- a. Wet vent
 - c. Unit vent
 - b. Back vent
 - d. Loop vent
88. A pipe that collects discharge direct from the plumbing fixture?
- a. Horizontal branch drain pipe
 - c. Branch vent
 - b. Fixture drain
 - d. Leader
89. A vertical pipe extended above the connection of horizontal branch drain?

- a. Vent stack
 - b. Soil stack
 - c. Stack vent
 - d. Vent stack thru roof
90. In water supply lines, it conveys water from building main to fixture groups in various location?
- a. Fixture supply pipe
 - b. Principal branch
 - c. Building Main
 - d. Riser pipe
91. It is usually used in a high rise building in regulating water pressure in pipe?
- a. Pressure relief valve
 - b. Globe valve
 - c. Check valve
 - d. Gauge valve
92. If vertical pipe that collects rain water from roof is located inside the building, it is called?
- a. Downspout
 - b. Leader
 - c. Gutter
 - d. Stack pipe
93. In a G.I. pipe connections, what fitting is used in connecting a pipe to an existing fixed pipe?
- a. Coupling
 - b. Union patente
 - c. Nipple
 - d. Reducer coupling
94. At what deep the P-trap crown be installed from finished floor line?
- a. 30 cm
 - b. 40 cm
 - c. 45 cm
 - d. 50 cm
95. A receptacle that collects water from surface and down spout?
- a. Storm sewers
 - b. Catch basin
 - c. Area drain
 - d. Sump pit
96. It is a devise in drainage system that prevents the reversal of the flow of liquid?
- a. House trap
 - b. Back flow valve
 - c. P-trap
 - d. Drum trap
97. What is one of the general conditions for a good waste pipe installation?
- a. Right location of clean outs
 - b. Right size of pipes
 - c. Right slope of grade of pipe lines
 - d. All of the above
98. Determine the size of a sanitary house drain for plumbing installation that has 92 fixture units with a 2 % slope?
- a. 3ø dia.
 - b. 4ø dia.
 - c. 5ø dia.
 - d. 6ø dia.
99. What is the storm drain pipe size in a 600 s.m. roof area with a slope of 4 %?
- a. 75 mm dia.
 - b. 100 mm dia.
 - c. 125 mm dia.
 - d. 150 mm dia.
100. What should be the size of a horizontal branch drain pipe with plumbing installation 2-lavatories and 1-floor drain?

- a. 20 dia.
- b. 30 dia.

- c. 30 dia.
- d. 40 dia.

End of Pre-Board Exam
Congratulation!

References:

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6. Plumbing Design Manual, by Regaredo Rebanos
7. Dictionary of Architecture & Construction, by Cyril Harris
8. Advanced Plumbing, by Harry Slater & Lee Smith
9. Plumbing Installation and Design, by James A. Sullivan
10. Utilities and Plumbing, by George Salvan

Answers:

1. A	11. B	21. A	31. C
2. C	12. A	22. B	32. A
3. B	13. A	23. B	33. D
4. C	14. B	24. B	34. C
5. D	15. B	25. A	35. B
6. D	16. D	26. D	36. B
7. B	17. D	27. A	37. B
8. A	18. B	28. C	38. B
9. C	19. A	29. A	39. B
10. C	20. B	30. B	40. B
41. C	51. C	61. D	71. B
42. B	52. C	62. C	72. D
43. A	53. D	63. B	73. A
44. D	54. C	64. C	74. A
45. D	55. C	65. D	75. C
46. C	56. B	66. B	76. C
47. A	57. B	67. A	77. D
48. B	58. B	68. A	78. C
49. A	59. A	69. C	79. C
50. C	60. A	70. B	80. D
81. C	91. A		
82. B	92. B		
83. D	93. B		
84. B	94. A		
85. B	95. B		
86. C	96. B		
87. B	97. D		
88. B	98. C		
89. C	99. B		
90. B	100. A		

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